

## Cover Letter

Submission to *Physica Scripta* as a Technical Note

*A Correlation-Production Reformulation of the  
Transfer-Entropy Second Law of Information Thermodynamics*

Dear Editor,

I am pleased to submit the manuscript entitled “A Correlation-Production Reformulation of the Transfer-Entropy Second Law of Information Thermodynamics” for consideration as a Technical Note in *Physica Scripta*.

This manuscript presents a concise technical reformulation of the generalized second law of information thermodynamics originally derived by Ito using transfer entropy and reversed transfer entropy. Instead of relying directly on the detailed fluctuation theorem, the manuscript derives the same bound from exact information-dynamical identities involving entropy, mutual information, transfer entropy, reversed transfer entropy, and explicit system-bath correlations. In this way, the work clarifies the relationship between the transfer-entropy formulation of information thermodynamics and the correlation-production approach to entropy production.

The main contribution is a transparent information-theoretic derivation connecting two existing approaches. Under separability, non-synergistic bath-coupling, large-bath, and deterministic reversible total-dynamics assumptions, the generalized second law is recovered while making explicit how system-bath correlations and dynamical assumptions enter the formulation. The manuscript also discusses how the resulting entropy production relates to the detailed-fluctuation-theorem expression in weak-coupling or high-temperature regimes, and why the correlation-production formulation may be useful when backward-path definitions become less transparent.

I believe that the manuscript is suitable for *Physica Scripta* as a Technical Note because it provides a compact theoretical clarification at the interface of statistical physics, information theory, and information thermodynamics. The work is self-contained, technical in nature, and intended to help organize existing formulations of entropy production and information flow within a unified information-dynamical framework.

The manuscript is original, has not been published elsewhere, and is not under consideration by another journal. No funding was received for this work, and I declare no competing interests.

Yours sincerely,

Yunju Lee